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Subject code & Name :	ITUE301: Advanced Web Development Frameworks	Practical :	8	Academic Year :	2026-27

Task - 8

Aim

To optimize the performance and bundle delivery of a React web application using code-splitting via React.lazy() and Suspense, isolating component re-renders with React.memo(), memoizing expensive operations with useMemo() and useCallback(), and configuring Vite manual chunk splitting.

Code

GitHub Link / Repository Link: https://github.com/Dharmikvaghani/AWDF-PRACTICAL.git

Key Implementation Highlights:

1. Lazy loaded components (TaskAnalytics) dynamically imported on demand with skeleton fallback.
2. TaskItem wrapped in React.memo with internal edit state to eliminate full-list re-renders.
3. Vite manualChunks splitting vendor-react and vendor-router into independent cached bundles.

```
// frontend/src/pages/TaskManager.jsx
const TaskAnalytics = lazy(() => import('../components/TaskAnalytics'));
// vite.config.js (Chunk Splitting)
build: {
  rollupOptions: {
    output: {
      manualChunks(id) {
        if (id.includes('node_modules/react') || id.includes('node_modules/react-dom')) return 'vendor-
react';
        if (id.includes('node_modules/react-router')) return 'vendor-router';
      }
    }
  }
}
Output
```

Output

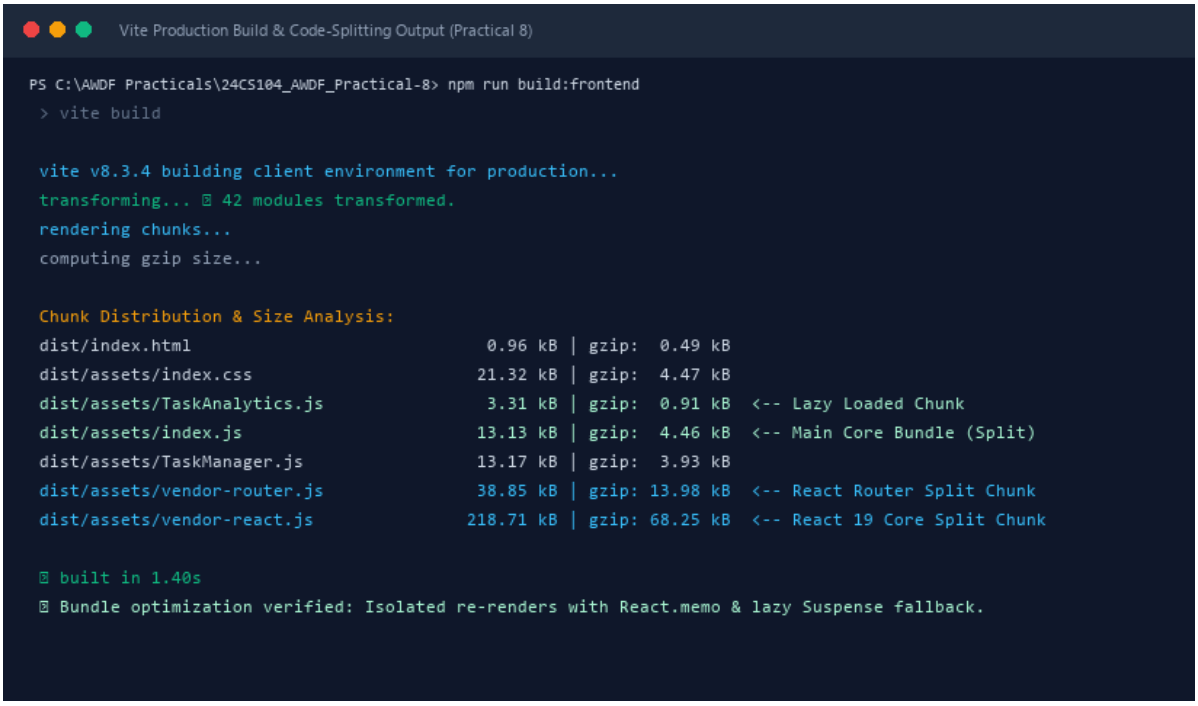


Figure 8: Execution Output & Functional Verification for Practical 8